

Jiayi (Johnny) Li

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EDUCATION

University of Toronto

09/2025 - 08/2026

HBSc in Computer Science & Quantitative Biology

Toronto, ON

- GPA: 4.0/4.0
- Relevant Coursework: Calculus with Proofs, Foundations of Comp Sci I & II, Writing Reports, Statistical Reasoning
- Activities: Biotechnology Innovation and Development, Science Communications, UofT Tri-Campus Hockey

RESEARCH EXPERIENCE

Radiation Dosimetry and Quality Assurance (QA) Research

01/2026 - Present

Undergraduate Researcher, Sunnybrook Hospital, under Dr. Arman Sarfehnia

North York, ON

- Conducted research on radiation dosimetry and quality assurance in radiotherapy, working toward ~1% precision for absolute reference dosimetry used to verify clinical treatment delivery
- Assisted in the design, testing, and evaluation of calorimetry-based detectors by characterizing response across multiple conditions and contributing to refinements that reduced measurement uncertainty
- Developed a Python program (NumPy, pyvisa) that automates temperature analysis via linear regression on pre/post-irradiation voltage drifts cutting per-dataset analysis from ~30 minutes to under a minute
- Built an acquisition system that samples up to 6 RTD probes on a multimeter at intervals, applying the Callendar–Van Dusen equation with per-probe calibration constants to deliver readings with live logging

Kidney Genetics and PAX2-Mediated Repair Research

11/2025 - Present

Undergraduate Researcher, University Health Network, under Dr. Moumita Barua

Toronto, ON

- Independently reanalyzed podocyte mass spectrometry data to validate P3H2-mediated type IV collagen modifications in Alport syndrome mouse models, cross-referencing 2 prior analyses for proteomic reproducibility
- Performed podocyte cell culture and mass spectrometry workflows, and conducted Western blot to validate Col4a3 antibody specificity on podocyte protein extracts for basement membrane integrity research
- Independently designed and deployed the Barua Lab's first-ever lab website using HTML, CSS, JavaScript, and Vue.js, publicly showcasing the lab's research, open datasets, and open-source software
- Selected as a funded IMS SURP fellow through the University of Toronto's Institute of Medical Science; presented research findings via poster at SURP Research Day

Cytomegalovirus (CMV) and Environmental Metal Exposure Research

03/2025 - 04/2026

Student Researcher, Northwestern University, under Dr. Hui Zhang

Remote, ON

- Analyzed 4 heavy metal biomarkers across ~554 U.S. adults from NHANES 2017–2018 to quantify associations between environmental metal exposure and persistent CMV infection
- Applied and built 5+ statistical models in R including multivariable regression, GAMs, and sensitivity analyses across all metal-CMV exposure-outcome pairs to control for confounders
- Performed 4 sensitivity analyses and Weighted Quantile Sum (WQS) mixture analysis using R packages including gWQS, survey, and ggplot2 across all exposure-outcome pairs
- Sole-authored research paper peer-reviewed and published by Asia Pacific Science Press in the *Advances in Management and Intelligent Technologies (AMIT)* journal

Parkinson's Disease (PD) and Protein Aggregation Research

07/2024 - 02/2026

Student Researcher, Harvard Medical School, under Dr. Ulf Dettmer

Boston, MA

- Performed 3 IncuCyte live-cell assays at Brigham & Women's Hospital, quantifying α -synuclein aggregation across 5 compounds, 3 doses, and 35+ replicates per condition
- Conducted quantitative image analysis of inclusion burden, YFP signal, and cell density across 500+ data points to identify small molecules that disaggregate lipid-rich α S inclusions
- First-author peer-reviewed publication in the *Journal of High School Science (JHSS)*; selected and presented a poster at the 2026 American Association for Advancement of Science (AAAS) Annual Meeting
- Delivered an oral talk and presentation at the 2025 CCIR Student Symposium to University of Cambridge faculty and Nobel Laureate Thomas R. Cech

PUBLICATIONS

- Li, J. (2026). Cytomegalovirus infection in population samples: Are whole-blood levels of cadmium, mercury, selenium, and manganese associated with CMV serostatus? *Advances in Management and Intelligent Technologies*, 2(1). <https://doi.org/10.62177/amit.v2i1.1094>
- Li, J & Dettmer, U. (2025). Parkinson's Disease-linked protein α -synuclein: do small-molecule inhibitors of protein aggregation also prevent its accumulation with lipids?. *Journal of High School Science*, 9(1). Retrieved from <https://doi.org/10.64336/001c.131827>
- Han, Z., Li, J., Yang, S., ... Shi, J. (2026). *Towards a nonlinear future: How to maximize personal value and achieve true growth in the age of AI*. China Financial & Economic Publishing House.

WORK EXPERIENCE

My Learning Space AI (MYLS)

06/2025 - 05/2026

AI Product Manager Intern

Toronto, Canada

- Conducted A/B testing and user interviews to identify key friction points, designed and iterated an optimized booking flow improving successful bookings by 9%, and drove data-backed improvements in monetization
- Created AI-generated mock prototypes and collaborated cross-functionally with engineering and design to support early-stage user research and implement data-backed product improvements
- Co-authored a 5-chapter book published by China Financial & Economic Publishing House on AI's structural impact on education and institutional decision-making in North America

Pharmaron Beijing Co., Ltd.

05/2025 - 08/2025

Chemical Analysis Researcher

Beijing, China

- Operated Liquid Chromatography–Mass Spectrometry (LC-MS) to analyze over 50 pharmaceutical samples, processing and interpreting mass spectra using Excel for compound identification and purity assessment.
- Collaborated with 8 professional chemists to develop and synthesize compounds used in widely distributed medications, including cold treatments and acetaminophen (Tylenol).
- Prepared and delivered 10+ research presentations translating LC–MS drug composition data into insights for international clients.

PROJECTS

S&P 500 Market Crash Simulator – Python

- Led a 4-person team building a financial contagion simulator across 503 S&P 500 companies using a weighted graph of ~24,000 Pearson correlation edges ($r \geq 0.7$) from COVID-19 period log-returns
- Engineered a recursive contagion algorithm in Python attenuating crash impact across correlated neighbors, simulating user-defined crash scenarios with real-time market cap loss estimates
- Built a full-stack interactive web dashboard using Dash and Plotly, integrating yfinance historical data retrieval, NetworkX graph layout computation, and NumPy correlation analysis into a single deployable pipeline

Nucleus Independent Living Services - Relating Chronic Conditions to Services

- Led a 3-person team analyzing ~4,000 Nucleus client records across 6 chronic burden groups to quantify 12-month pre/post changes in hospital days, healthcare costs, and readmission rates in R
- Found clients with 5+ chronic conditions had a median 7-day hospital reduction and respiratory clients saw the largest average cost reduction at ~\$22,500, identifying highest-need client groups for Nucleus services
- Performed data wrangling across 3 research questions, engineering variables for chronic burden, cost reduction, and age groupings, producing summary tables and visualizations using ggplot2

SKILLS AND INTERESTS

- **Programming Languages:** Python, HTML/CSS, Javascript, R, Microsoft Offices, CAD, Latex
- **Frameworks/Libraries:** Flask, Vue, React, Dash, Plotly, PyVisa
- **Developer Tools:** Git, VS Code, PyCharm
- **Biomedical Research:** Electrophoresis, RNAi, PCR, CRISPR, Gradient dilution, Fluorescent Microscopy